

Estimating Treatment Effects in the Presence of Overlapping Programs and Misreporting*

Erica L. Mtenga[†] Pierre E. Nguimkeu[‡]

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Abstract

Non-mutually exclusive interventions are common features of programs implemented in developing and developed countries. Evaluating the impact of individual and joint participation in these programs may be of policy interest in guiding the cost-effective allocation of resources. However, program participation is substantially misreported in survey data, which may result in biased treatment effect estimates. While the literature has focused on misreporting in one program, our study proposes a method to consistently estimate individual and joint treatment effects of potentially misreported overlapping (and exogenous) programs. We focus on false negative cases which are more prevalent in observational studies. We derive the bias in the traditional ordinary least squares (OLS) estimator and show that it is not possible to determine the direction of the bias a priori. The joint treatment effect may also have an opposite sign to the true effect, which may have dramatic consequences if used to inform policy on whether the programs are complements or substitutes. As in the previous literature, we argue that any instrumental variable (IV) that meets relevant criteria fails to adhere to the exclusion restrictions, resulting in biased IV estimates. We then develop a consistent estimator of treatment effects using misclassification probabilities, available through validation studies and other external sources. When misclassification probabilities are unknown, we provide an approach to estimate and apply them in the proposed method. Monte Carlo simulations show that the estimator performs well in finite samples. Finally, we provide an empirical example, estimating the effect of two correlated and documented to be substantially misreported programs, the Supplemental Nutrition Assistance Program (SNAP) and the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), on food security and healthy eating.

JEL Classification: C13, C31, H53, I38

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[†]Corresponding Author. Department of Economics, Georgia State University, Atlanta, GA 30303; Email: emtenga1@gsu.edu

[‡]Department of Economics, Georgia State University, Atlanta, GA 30303; Email: nguimkeu@gsu.edu